

Analytics Summary

1. Average GRE score of an accepted PhD applicant

No valid GRE scores were available for accepted PhD applicants after cleaning, so an average GRE score could not be calculated. This reflects missing self-reported information rather than a processing error.

2. Average GPA of an accepted Master's applicant

No valid GPA values were available for accepted Master's applicants after cleaning. Consequently, an average GPA could not be calculated because no observations met the filtering criteria.

3. Limitations of inferring decision_date

Decision dates were inferred from free-text status values and the application term. This approach can introduce errors because status text is inconsistent, the application term may not match the actual decision year, and some records do not contain enough information to infer a date.

4. Correlation analysis

Pearson correlation analysis found almost no linear relationship between GPA and GRE-related measures. GPA versus GRE produced a correlation of approximately -0.026 ($p = 0.406$), while GRE versus GRE Verbal produced a correlation of approximately 0.010 ($p = 0.748$). Both relationships were weak and not statistically significant.

5. GPA comparison

A Welch two-sample t-test comparing accepted and rejected applicants found no statistically significant difference in GPA ($t = 1.65$, $p = 0.098$). Although accepted applicants had a slightly higher average GPA, the difference was small and could reasonably be explained by random variation.

6. Chi-square test

A chi-square test comparing degree type and applicant type (American versus International) produced $\chi^2 = 3.42$ with 1 degree of freedom and a p-value of 0.065. Because the p-value is greater than 0.05, there is insufficient evidence to conclude that these variables are associated in this dataset.

7. Remaining data quality issues

Missing GPA and GRE values remain common after cleaning. Some variables, including inferred decision dates, were derived from text rather than directly recorded, introducing uncertainty. Because the dataset is self-reported, inconsistencies and reporting bias may still be present.